

1-A:

A - This is a correct description of a positive externality. By vaccinating children, the medical team prevents the spread of the disease, which benefits others in the camp who may not have been vaccinated. This is an additional benefit that arises from the vaccination program.

B. The benefit described here is directly related to the children receiving medical help which represents a private benefit.

C. This option is irrelevant to the concept of positive externality. It highlights the importance of the medical team's intervention but doesn't describe the broader benefit to others.

D. If the social benefits are less than the social costs, it would be a net loss for society because there is existence of net social costs.

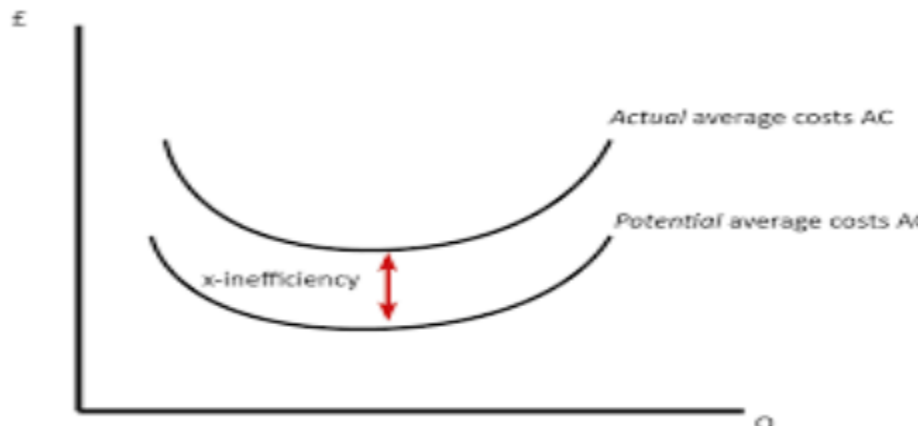
2-C:

Based on the diagram, if the firm moves from point X to point Y, the following changes are likely to occur:

Productive Efficiency:

- **Improves:** The firm is moving from a higher average cost curve (X) to a lower average cost curve (Y), indicating that it is becoming more efficient in its production process.

X-Inefficiency:



X-inefficiency refers to the inefficiencies within a company that result in higher production costs than necessary for a given output. These inefficiencies can stem from a variety of factors, including outdated technology, inefficient production processes, poor management, and a lack of competition.

- **Decreases:** X-inefficiency refers to the difference between the actual average cost (X) and the potential average cost (Y). By moving from X to Y, the firm is reducing its X-inefficiency.

3-A:

Socially optimal level of output explains a situation whereby marginal social benefits (additional benefits to the society) are equal to marginal social costs (additional costs to the society)

This explains why key A is correct.

The marginal cost of pollution does not have to be zero (option A). Secondly, if the total revenue equals to total cost, this represents a normal profit situation. (Option C). Moreover, there are no benefits from pollution (option D)

4-B:

A: The rational consumer will not wait until his marginal utility becomes zero; instead, as soon as he faces diminishing marginal utility, he will switch to more of good Y, to maximize his satisfaction.

$$\bullet \frac{MU_X}{P_X} > \frac{MU_Y}{P_Y} \quad (B)$$

B – If the price of Y rises, then X becomes relatively cheaper, and therefore, the rational consumer will switch the consumption of good Y to good X to maximize his satisfaction.

This is why, key B, will be correct.

C – In-order for the total utility to be negative, the marginal utility must be negative. The question says that the marginal utility of good X is positive but diminishing, which means that the total utility will be increasing but at a decreasing rate.

D – The question is saying that the marginal utility is falling, this means that the **total utility will be increasing at a decreasing rate.**

5-D:

An indifference curve represents various combinations of two goods that provide the same level of total satisfaction or total utility to a consumer. All points on the indifference curve represent the same total utility.

The indifference curve shows the **various trade-offs between two goods that leave the consumer indifferent**, meaning the consumer is equally satisfied with each combination.

This is why, a rational consumer is indifferent along the indifference curve.

This is why, all points E, F and G (which are on the same indifference curve) are equally desirable combinations since they yield the same total utility.

6-D:

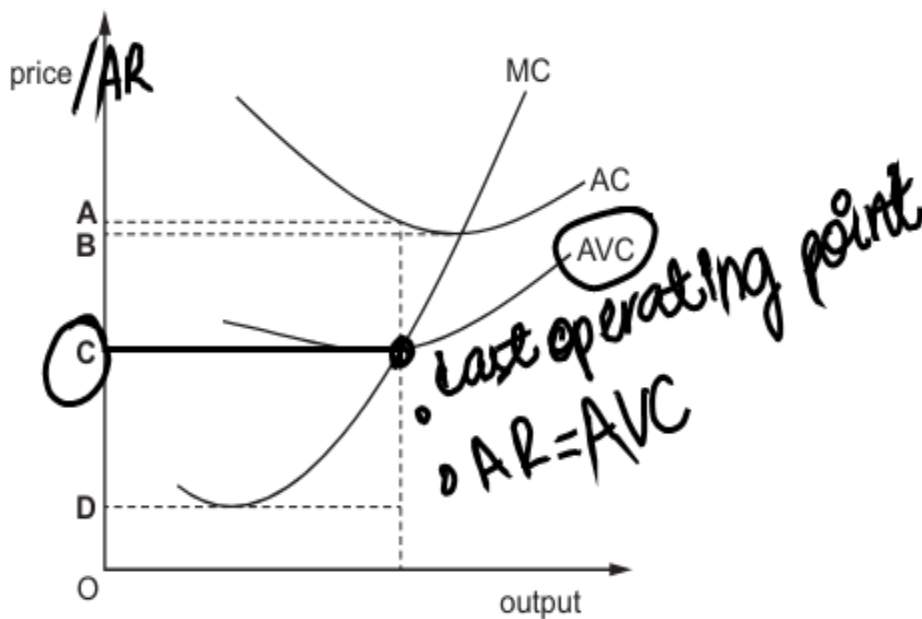
Short-Run Production:

In the short run, a firm's fixed costs remain unchanged, but variable costs increase as the firm produces more output. When a firm produces an extra unit of output:

- **Variable costs** increase (e.g., raw materials, labor, energy)
- **Total costs (sum of FC and VC)** increase since there is a rise in variable costs.
- **Fixed costs** remain the same (e.g., rent, salaries, insurance)

7-C:

In the short-run, the firm's last operating point is where its Average revenue covers its Average variable cost. Below this point, the firm will shut down in the short-run due to its failure to cover its variable costs that form the significant proportion of total costs in the short-run.



This explains why option C will be correct.

8-A:

The table shows a firm's total and marginal costs.

cost incurred at zero level of output is fixed cost

output	total cost (\$)	marginal cost (\$)
0	300	
1	340	40
2	375	35
3	400	25
4	435	35
5	475	40

fixed cost

$MC_1 = TC_1 - TC_0$
 $40 = 340 - X$
 $X \Rightarrow 300$

What is the average fixed cost of producing 6 units?

- A** \$50 B \$60 C \$180 D \$300

$afc = fc / output \Rightarrow 300 / 6 \Rightarrow 50$

9-C:

Internal Growth and Diversification:

The electronics firm has:

- Invested in research and development (R&D) to improve its existing products (semiconductors)
- Applied its expertise in semiconductors to enter new markets (televisions and mobile phones)

This is an example of internal growth, as the firm is expanding its product line and entering new markets using its own resources and capabilities. Additionally, the firm is diversifying its product portfolio by entering new industries.

There is no indication of external growth through mergers and acquisitions, and sales revenue maximization is not the primary focus in this scenario.

This explains why other options will be incorrect.

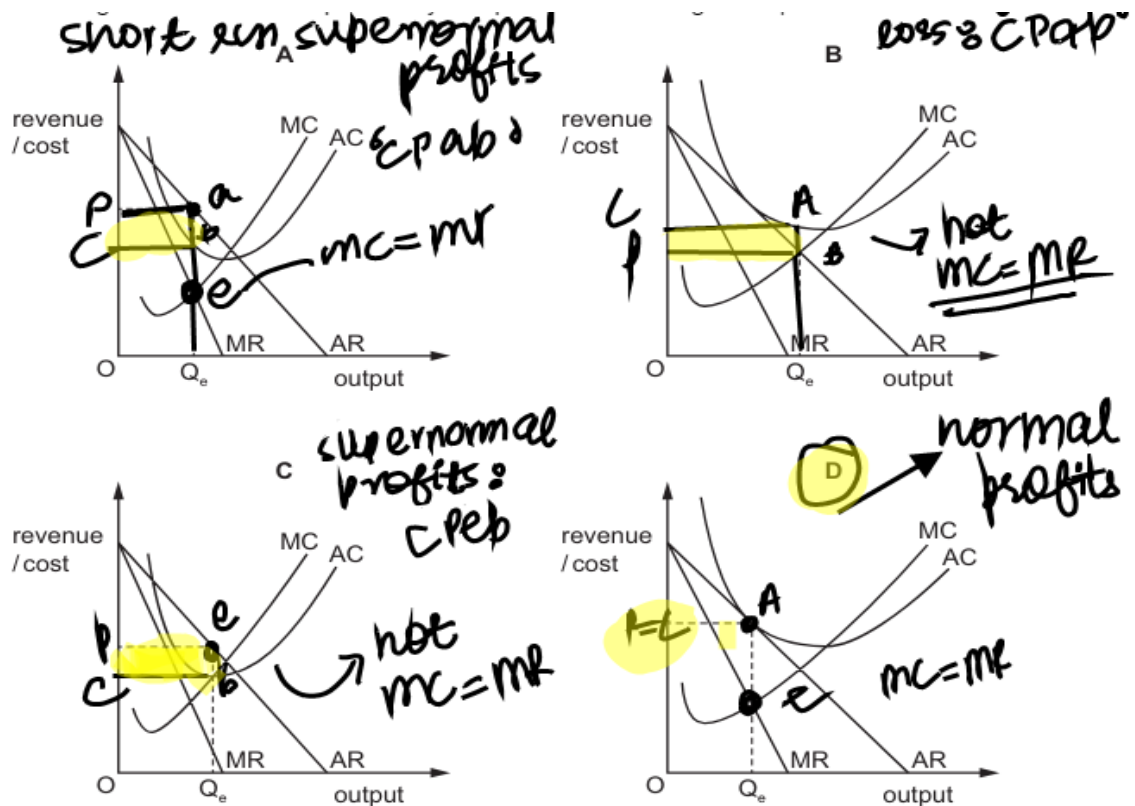
10-D:

A: This represents a short-run supernormal profits situation of a monopolistically competitive firm whose traditional objective is profit maximization since it operates where $MC=MR$.

B: A monopolistically competitive firm does not operate at any other situation apart from where $MC=MR$. Also, it does not make losses in the long-run.

C- A monopolistically competitive firm does not operate at any other situation apart from where $MC=MR$. Also, in the long-run , it does not earn any super normal profits.

D – This is correct because not only does a monopolistically competitive firm operate where $MC=MR$, but also, in the long-run, it earns normal profits or zero-economic profits where price(AR) is equal to Average total cost.



11-D:

A cartel will find it difficult to fix prices in the following scenarios:

1. **Low membership:** When the proportion of firms in the industry that are members of the cartel is low, it's challenging to control prices because the cartel's output reduction to increase prices may be offset by non-member firms increasing their production, and when overall market output rises, and market prices fall, the agreement may weaken.
2. **Low barriers to entry:** When barriers to entry are low, new firms can easily enter the market, making it difficult for the cartel to maintain price fixing. **This is because the market may become contestable, and the threat of potential competition may mean that firms cannot collectively act as a monopoly through a cartel agreement.**

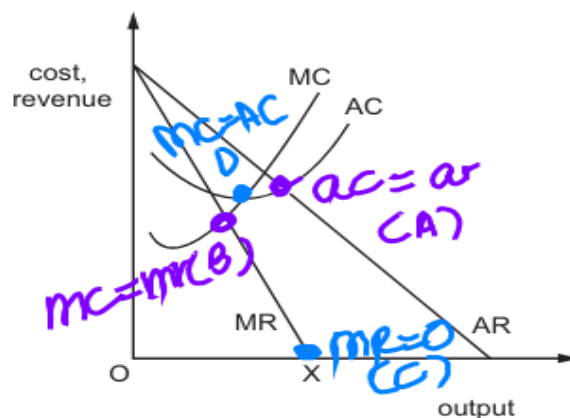
12-C:

A- Normal profits will be achieved when Average revenue is equal to Average total cost. This is shown through point A in the diagram below.

B – Total profit is maximized when the firm produces at the profit maximizing level of output where Marginal cost is equal to Marginal revenue. This can be seen through point B

C – Total revenue is maximized where Marginal revenue is equal to zero. This can be seen through point C. At output level of OX, total revenue will therefore be maximized. This explains why option C will be correct.

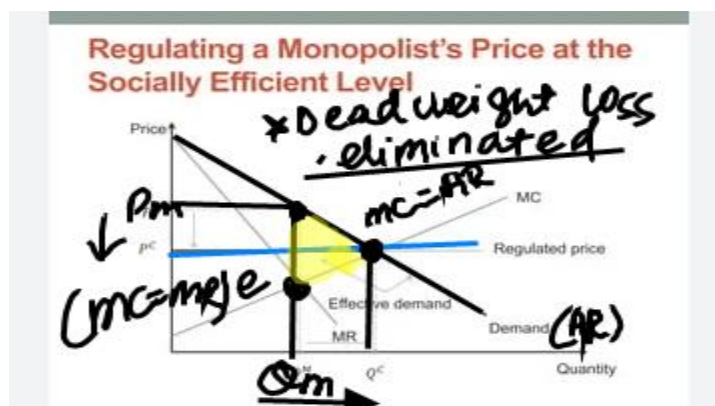
D – Average cost is minimized when Marginal cost is equal to Average total cost. This productively efficient point can be seen through point D.



13- D:

A – Government involving itself in direct provision of healthcare will help to solve the market failure associated with under-allocation of merit goods.

B – Price controls on large firms, such as: a monopoly firm, whereby a monopoly is bound to charge socially desirable level of price where $MC=AR$, will help to solve the market failure associated with monopolistic exploitation.



C – Regulations to limit smoke pollution will help to overcome market failure associated with over-allocation of demerit goods.

D – Removing import quotas does represent a move towards free trade as part of globalization, but it has no direct role in solving market failure. This explains why key D is correct.

14- B:

A – In the case of chemical firm, which imposes “toxic fumes”, external costs are generated, and no external benefits are generated.

B -

A chemical firm produces toxic fumes that impose costs on society. The diagram shows the free market equilibrium of the firm's product at P_1, Q_1 .



This is correct because when **tax that represents external cost, is included in the marginal private cost, it will help to obtain marginal social cost.** This way, the external cost or the negative externality is internalized, because we are making the party responsible for the harm that they are causing.

C and D: Because of imposition of a tax, the private benefits or the private costs are not internalized, because they are already considered by the private firm.

15-C

A - Means-tested benefits are not determined by age. This is because if the government provides benefits to **all elderly** in form of pension support , this will be considered as a “universal benefit”

B – If family size is considered, this represents universal benefits.

Suppose the government announces a benefit program: "Every family with four or more children will receive a monthly stipend of PKR 10,000."

In this scenario:

- **Family A has four children, but their income is PKR 500,000 per month. They don't need the benefit.**

- Family B also has four children, but their income is PKR 50,000 per month. They need the benefit.

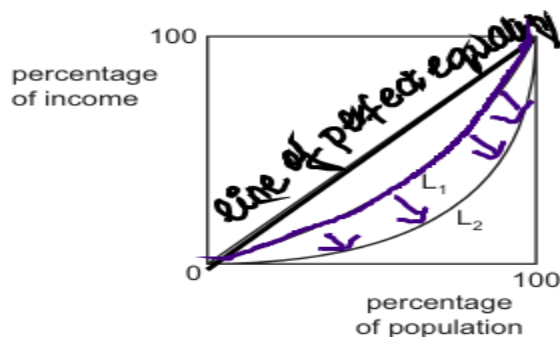
In this case, both families receive the benefit (PKR 10,000) solely because they have four or more children, regardless of their income level. This is a universal benefit, not means-tested.

C - Means-testing would consider the income level of each family and provide assistance only to those who need it most. This explains why key C will be correct.

D - Qualifications (e.g., education, skills) are not a primary factor in determining means-tested benefits.

16-B:

Lorenz curve represents the graphical representation of income distribution among a given population, which then helps us to gain insights on the level of inequality in an economy.



In this graph, the Lorenz curve is seen to further diverge away from the line of perfect equality from L1 to L2, which represents that inequalities have increased.

A, C: Both, capital gains tax and inheritance taxes are examples of progressive taxes. If these are reduced, this would mean that income inequalities may rise. This would be "most likely to cause this shift"

D - A tax-free allowance, also known as a personal allowance or basic exemption, is the amount of income that an individual is allowed to earn without paying income tax. It is a threshold set by the government, below which income is not subject to taxation.

In other words, it is the amount of income that is "tax-free" or exempt from income tax. Any income earned above this allowance is subject to income tax.

For example, if the tax-free allowance is PKR 50,000, an individual with an income of PKR 40,000 would not pay income tax, while an individual with an income of PKR 60,000 would pay tax on the amount above PKR 50,000 (PKR 10,000)

If the tax free allowance has been reduced for everyone, this means that those with lower incomes who rely more heavily on the tax-free allowance to make ends meet will face problems as it would proportionally affect them more, widening the income gap between

the rich and the poor. With a reduced tax-free allowance, lower-income individuals may be pushed into higher tax brackets, increasing their tax burden and reducing their disposable income. **This is why, this will also be most likely to cause this shift.**

B – If the income tax system is made more progressive which represents the fact that high income individuals now pay a greater proportion of their incomes in taxes, this will result in higher tax revenues, which the government can utilize to effectively redistribute incomes from the high income groups to the lower income groups by means of means tested benefits. This explains why **this will be least likely to cause this shift, as this instead proves to reduce inequalities.**

17-B:

In a perfectly competitive market, the firm is a price-taker, and the marginal revenue (MR) is equal to the market price (P). This explains that **the average revenue and marginal revenue** will be the same, making these options incorrect.

The marginal revenue product (MRP) of a factor of production is the additional revenue generated by employing one more unit of the factor.

As the firm employs more of the factor, the marginal physical product (MPP) of the factor falls due to the law of diminishing returns. Since MR is constant (equal to the market price), the MRP (which is the product of MR and MPP) also falls as MPP falls.

Here's the formula to illustrate this:

$$\text{MRP} = \text{MR} \times \text{MPP}$$

Since MR is constant, a fall in MPP leads to a fall in MRP.

18-C:

If labor migrates from developing countries to developed countries, this is most likely to **increase the size of the labor force in the developed countries, because of which, there will be potential economic growth in these countries.**

Moreover, this will also result in a surplus of labor in developed countries that will bring about a downward pressure on wage rises **or in other words, wages will decrease.**

This explains why option C will be correct.

19-D:

A. Absolute advantage is a key concept in trade theory. This refers to the ability of a country to produce more goods or services than another country, using the same amount of resources (such as labor, capital, and technology).

B. The allocation of resources based on comparative cost advantage is a fundamental principle of free trade, where countries specialize in goods where they have a lower opportunity cost relative the other country.

C. The international division of labor and differences in opportunity cost ratios are also central to trade theory, as countries specialize in goods based on their comparative advantage.

D: -Free trade models assume a simplified scenario where goods and services can be traded without any costs or barriers. However, in reality, trade is often subject to various costs and inefficiencies, such as:

- Transport costs (e.g., shipping, logistics)
- Inefficiencies in handling goods and services (e.g., bureaucracy, customs clearance)
- Differences in standards and regulation

20- C:

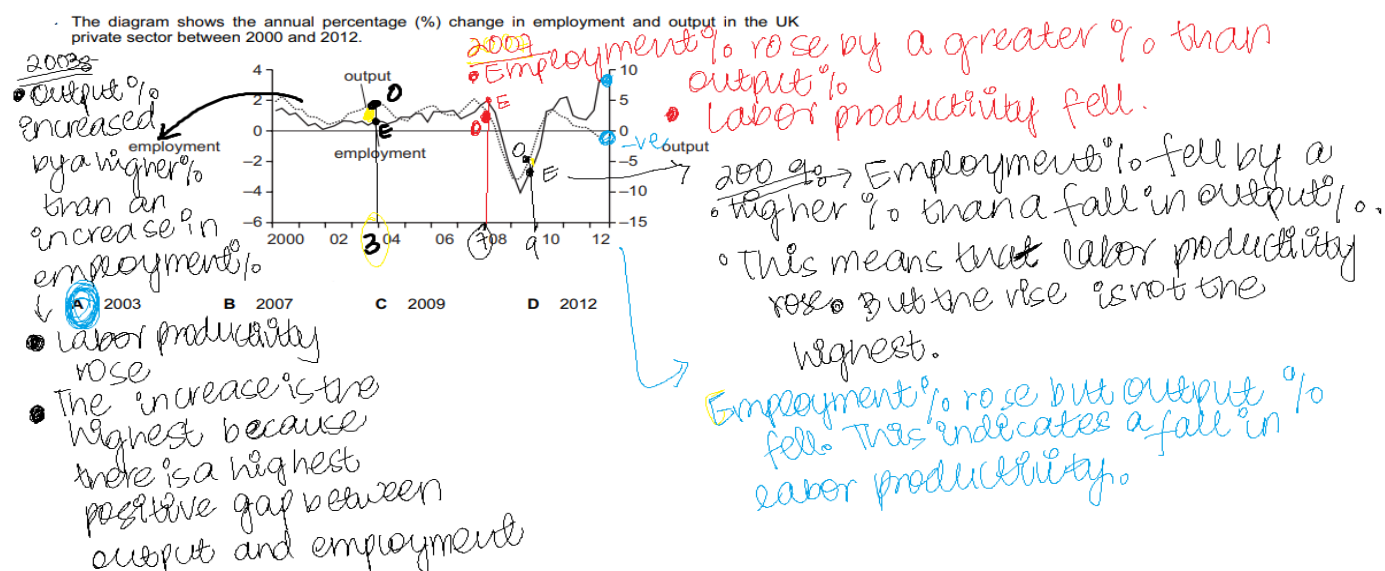
A – If there is already a shortage of vacancies, then the government policy to provide transfer payments to those seeking work and providing information on vacancies, would not contribute greatly to employment.

B – Since no production of goods and services is required for a transfer payment, this means that there would be no contribution to gdp, which is a negative outcome of this policy.

C – Availability of information on jobs available distinguishes a perfect labor market from an imperfect one, which means that the government's attempts to provide information on jobs available will make the labor market more perfect. This is why, key C, will be correct.

D- If students leave their studies to join the workforce at an early age, the overall quality of human capital will decline, which represents a negative outcome of this policy.

21-A:



In order to check the year in which labor productivity increased the most , we will need to check the labor productivity at the end of the year.

The formula for labor productivity is:-

Output/Employment

For labor productivity to increase by the most, output should increase by a greater proportion than employment. It can be only be seen in 2003 where the rate of growth of output exceeded the rate of rise of employment by the highest percentage.

22-B:

22 The table shows the levels of consumption expenditure and savings for given family incomes.

	$\Delta \text{in } Y$	disposable family income (\$)	consumption expenditure (\$)	savings (\$)
A	1000	2000	2150	-150
B	1000	3000	3100	-100
C	1000	4000	4000	0
D	1000	5000	4850	150
E	1000	6000	5650	350
		7000	6380	620

$$MPC: \frac{\Delta \text{in } C}{\Delta \text{in } Y}$$

$$A: \frac{950}{1000} \rightarrow 0.95$$

$$B: \frac{900}{1000} \rightarrow 0.90$$

$$C: \frac{850}{1000} \rightarrow 0.85$$

$$D: \frac{730}{1000} \rightarrow 0.73$$

falls continuously

Over the range of disposable income shown, as income rises the marginal propensity to consume

- A falls and then rises.
- ☒ B falls continuously.
- C rises and then falls.
- D rises continuously.

23-B:

Change in real income per head (%) => change in nominal income (%) - change in prices (%) - change in total population (%)

$$\rightarrow 3\% - 4\% - 1\%$$

$$\rightarrow -2\% \text{ (Real income per head decreased by 2\%)}$$

24- C:

A – How changes in consumption lead to changes in income does not represent the accelerator principle.

B – When changes in income lead to changes in consumption , this represents marginal propensity to consume.

C – According to accelerator theory, investment is a function of a rate of change of income. If the change in real national income increases, then induced investment will increase as firms increase the demand for capital goods to meet increases in demand.

➤ The formula to find the co-efficient of accelerator is :

Change in Investment/ Change in Income

Case I. Life of the Machine 10 years							
	Period	Change in consumption	Capital Equipment needed	Additions	Replacement	Total	Percentage change in Gross Investment
10%	0	1000	100	Nil	10	10	—
rise in demand	1	1100	110	10	10	20	100% increase

Handwritten notes: "autonomous investment" with an arrow pointing to the 'Total' column; "induced" with a circle around the '10' in the 'Additions' column for period 1.

D – When changes in investment result in changes in income, this represents the multiplier process.

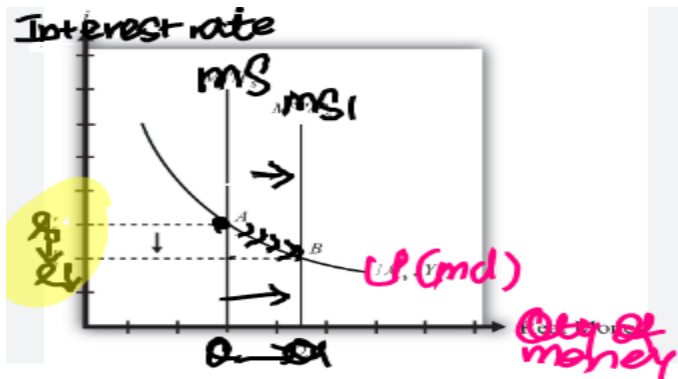
25-A:

- A- When the frequency of income payments increases, for example: when there is a switch made from monthly payments to weekly payments of wages by employers, individuals receive their income more frequently, reducing their need to hold money for transactionary purposes for long periods. This leads to a decrease in the demand for money, causing the demand curve to shift to the left.
- B- When the general price level rises, or in other words, there is inflation, the real value of money will erode. This means that in order to buy the same amount of goods and services, individuals will require more money since the purchasing value of money has fallen, which will further lead to an increase in demand for money.
- C- If incomes rise, this will result in a rise in the transactionary demand for money, which is strongly linked with changes in incomes.
- D- If the rate of interest rises, this will lead to a contraction in the quantity demanded of money, or in other words, there will be a movement along the liquidity preference curve instead of a leftward shift.

26-B:

If the central bank performs **quantitative easing, i.e. it buys back government securities, for example: long term bonds from the non-bank private sector** on behalf of the government, this will increase money supply of the economy, as now stocks of narrow money and broad money will rise.

When money demand or liquidity preference is unchanged, this will result in a surplus of money in the money market, as a result of which, the price of money or the interest rates will fall. This can be seen in the diagram below.



27-C:

The correct answer is C: There is a need for individual choice expressed through the market.

Both Keynesians and Monetarists believe in the importance of individual choice and the role of markets in allocating resources. They both support the idea that individuals should be free to make their own economic decisions. This explains why key C is correct.

A. Keynesians and Monetarists have different views on expectations. Keynesians believe that expectations may not adapt rapidly to changing circumstances. Monetarists, on the other hand, believe that expectations adapt quickly to new information.

B. Keynesians believe that fiscal policy (government spending and taxation) is an effective tool for stabilizing the economy, while Monetarists believe that monetary policy (central bank control of interest rates and money supply) is more effective.

D. Monetarists believe that the velocity of circulation of money (the rate at which money is spent and re-spent in the economy) is stable, but Keynesians do not share this belief. Keynesians argue that velocity of circulation is unstable.

28-C:

Keynesians believe that the economy is subject to sticky prices and wages, which can lead to deviations from full employment even in the long run. They argue that these imperfections can cause the economy to remain below full capacity for an extended period, requiring government intervention to stabilize the economy. This explains why key C is correct.

A. Keynesians do not believe that deviations from full employment are always temporary. Instead, they argue that the economy can remain in a state of unemployment for a long time

without self-correcting. **Instead, monetarists believe that the economy will always achieve full employment in the long-run.**

B. Keynesians do not believe that free markets always lead to efficient outcomes. Instead, they argue that market imperfections can lead to inefficiencies, which is why, they believe that government intervention in form of expansionary fiscal policy is required to stabilize the economy.

D. Keynesians believe that expansionary fiscal policy (government spending and taxation) can play a significant role in stabilizing the economy and addressing economic downturns, contrary to the statement.

29-A:

Because of quantitative easing by the central bank on behalf of the government, that is when, it buys back previously issued securities from the non-bank private sector, the money supply increases, as a result of which, interest rates fall.

Note: External value of the currency can be assessed by observing the impact on the exchange rate. Additionally, internal value of the currency represents its real value.

- When interest rates fall, foreign investors who invested in assets in the country may withdraw their funds to avoid losses or seek better returns elsewhere. This leads to a sudden outflow of capital, known as hot money outflows. Because of this, the demand for domestic currency may fall, which results in depreciation of the exchange rate. **This represents that there is a fall in the external value of a country's currency in terms of another currency.**

->According to the Monetarists' Quantity Theory of Money ($MV=PT$), an increase in the money supply (M) leads to an proportional increase in the price level (P), assuming the velocity of money (V) and the volume of transactions (T) remain constant.

This means that if the money supply increases by a certain percentage, the price level will also increase by the same percentage. As a result, the internal value or real value of the currency will fall by the same percentage.

This explains why key A will be correct.

30-B:

A - The tariff makes US steel exports less competitive, as other countries may retaliate with their own tariffs.

B – Because of an increase in protectionism by means of tariff, the competitive pressures faced by domestic steel industries will decrease. This might translate into a loss of productive efficiency as domestic firms choose to be X-inefficient since now there may be no incentive to work hard. (Potential for domestic producers to reduce innovation and investment, as they may rely on protection rather than competitiveness)

C. Terms of trade is calculated using: $\text{export prices} / \text{import prices} * 100$. While tariffs raise import prices, which can actually worsen terms of trade, we don't know what is happening to export prices to deduce the net impact on terms of trade.

D. The tariff increases the producer surplus for US steel producers, as now they sell a higher quantity of steel at a higher price. This can be seen through the region “1” in the diagram below.

